

# Carry-lookahead G/P

Ripple-carry waits for each Cout before the next Cin

# Five plus three starter

- Starter: four-bit A equals five, B equals three, C zero zero
- Bit zero: both inputs one
- Step to reveal C one, C two
- Full reveal: sum eight, Cout zero
- If G is one, carry out is one regardless of Cin
- If G is zero and P is one, carry passes

# Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

Carry-lookahead generate & propagate

See  $G_i = A_i \cdot B_i$  and  $P_i = A_i \oplus B_i$ , then expand carries in parallel:  $C_{i+1} = G_i + P_i \cdot C_i$ . Starter: 4-bit  $5 + 3 = 8$ .

Starter example: 4-bit  $5 + 3$ . Read G/P per bit, then reveal  $C_1, C_2, \dots$  from the expanded lookahead equations.

Challenge 0 / 22 cleared

Quiz: generate: Generate  $G_i$  is defined here as...

☐  $A_i \cdot B_i$

☐  $A_i \oplus B_i$

☐  $A_i + B_i$

☐  $C_i$  alone

Show hint

Check

Next

Load challenge setup

Idle

Quiz: generate

Quiz: propagate

Quiz: recur carry

Quiz: sum

Quiz: why CLA

Quiz: G=1

Quiz: P pass

Quiz: kill

5 + 3

See G0

See P1

C1 from G0

15 + 1

C0 = 1

8-bit 100+50

All generates

XOR ones

Expand C2

Quiz: OR propagate

Quiz: vs RCA

0 + 0

Reveal all

Carry-lookahead starter

learn\_digital — carry look ahead adder propagate and generate

# Workbook practice

- In the workbook track, write G, P, and C one for bit zero of five plus three
- Expand C two symbolically from G one, P one, G zero, and C zero
- Explain generate versus propagate in one sentence each
- Sketch why CLA trades area for speed versus ripple
- Name one pitfall: confusing G with P or forgetting sum uses  $P \oplus C$

# Pitfalls to watch

- Do not treat expanded equations as magic, each term is a real AND-OR path
- $G$  equals one always wins;  $P$  equals zero kills carry regardless of  $C_{in}$
- Wide adders use prefix trees, not one giant flat expansion
- And remember: the browser lab is literacy
- Real RTL still needs sensible grouping and timing on carry paths

# Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, finish a few challenges after the starter
- On paper, write one recursive C equation and one expanded C two
- When you are ready, take the short quiz, then continue to the array multiplier